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Analysis of the Pollutant Formation in the FLOX® Combustion

FLOX®, or flameless combustion is characterized by ultralow NO_x *emissions. Therefore the potential for its implementation in gas turbine combustors is investigated in recent research activities. The major concern of the present paper is the numerical simulation of flow and combustion in a FLOX®-combustor [Wünning, J. A., and Wünning, J. G., 1997, "Progress in Energy and Combustion Science,"* ***23****, pp. 81–94; Patent EP 0463218] at high pressure operating conditions with emphasis on the pollutant formation. FLOX®-combustion is a highly turbulent and high-velocity combustion process, which is strongly dominated by turbulent mixing and chemical nonequilibrium effects. By this means the thermal nitric oxide formation is reduced to a minimum, because even in the nonpremixed case the maximum combustion temperature does not or rather slightly exceeds the adiabatic flame temperature of the global mixture due to almost perfectly mixed reactants prior to combustion. In a turbulent flow, the key aspects of a combustion model are twofold: (i) chemistry and (ii) turbulence/chemistry interaction. In the FLOX®-combustion we find that both physical mechanisms are of equal importance. Throughout our simulations we use the complex finite rate chemistry scheme GRI3.0 for methane and a simple partially stirred reactor (PaSR) model to account for the turbulence effect on the combustion. The computational results agree well with experimental data obtained in DLR test facilities. For a pressure level of 20 bar, a burner load of 417 kW and an air to fuel ratio of* $\lambda = 2.16$ *computational results are presented and compared with experimental data.* [DOI: 10.1115/1.2747266]

Keywords: FLOX®-combustion, numerical simulation, complex finite rate chemistry, PaSR model

Introduction of the Flox Principle

FLOX® is a registered trademark of WS Wärmeprozesstechnik [1]. The combustor design was supplied by this company and the manufacturing, particularly with regard to the specific demands of the high pressure operation, was accomplished by DLR. The term FLOX® is an acronym for "flameless oxidation," which actually means that no distinct flame is visible. The way in which this is performed and what follows as a consequence thereof, will be pointed out in the following. The fuel used in the experiments and the simulations is natural gas and its composition is depicted in Table 1.

In order to give a first impression of the FLOX® principle a longitudinal slice through combustor axis and two subtending nozzles is displayed in Fig. 1. The streamlines represent the flow and we recognize an outer and a stronger inner recirculating vortex that stabilizes the flame. In view of the pollutant formation the molar fraction distribution of CO and NO is shown on the upper and lower half of Fig. 1, respectively. At first glance we observe that the reaction zone is far detached from the nozzle exit.

In principle, FLOX® is a technique employed to reduce nitric-oxide (NO) emission to a minimum within the combustion process. The relevant NO-formation mechanisms during methane combustion are: (i) Zeldovich (thermal NO), (ii) Fenimore (prompt NO), and (iii) N_2O-NO. It is generally accepted that with increasing combustion temperature the thermal NO-formation becomes absolutely dominant over the other mechanisms. Therefore, low NO-emission values may be attained if the combustion is controlled in a manner in which the temperature does not exceed the adiabatic flame temperature of the global mixture.

Equilibrium computations reveal that increasing adiabatic flame temperature correlates with increasing preheating temperature and with decreasing air to fuel ratio λ. In order to maintain high operational efficiency, the preheating temperature has to be as high as possible. Consequently, adiabatic flame temperatures below 1800 K imply an air to fuel ratio λ in the range of $2 < \lambda < 3$.

If reaction occurs before the mixture is globally mixed on the molecular level, the local flame temperature is higher than the adiabatic flame temperature of the global mixture due to regimes of lower air to fuel ratio which are closer to stoichiometry than the global mixture. This should be avoided, in particular, flashback into the nozzle is absolutely prohibitive.

As is seen in Fig. 2, the mixing of fuel and oxidizer taking place in the burner is not complete; indeed, there is a slight non-concentric distribution of methane on the nozzle exit cross section with a maximum concentration of 17.7% near the nozzle axis and a minimum concentration of 0.3% at the nozzle walls. Details of the nozzle design are not shown here.

It will be seen in the computational results that this incomplete mixing in the burner is of less relevance in the FLOX®-combustion. The key is that mixing proceeds at extremely high flow velocity. The consequence is that on the one hand turbulent mixing is highly improved in the shear layer around the impinging jet and on the other hand that chemical reactions are strongly delayed by reason of nonequilibrium effects, even for high preheating temperatures. This is also the reason why simulations using combustion models on the basis of equilibrium or near-

Contributed by the International Gas Turbine Institute of ASME for publication in the Journal of Engineering for Gas Turbines and Power. Manuscript received January 4, 2007; final manuscript received January 29, 2007, published online January 9, 2008. Review conducted by Dilip R. Ballal. Paper presented at the ASME Turbo Expo 2006: Land, Sea and Air (GT2006), May 8–11, 2006, Barcelona, Spain, Paper No. GT2006-91041.

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Table 1 Operational conditions. The air to fuel ratio is defined as $\lambda=(n_{Air}/n_{Fuel})/(n_{Air}/n_{Fuel})_{stoic}$, where n is the respective number of moles and the subscript stoic denotes stoichlometry

Air to fuel ratio λ of the global mixture	(1)	2.16
Pressure	(bar)	20
Air temperature	(K)	721
Fuel temperature	(K)	352
Air mass flow	(g/s)	302
Fuel mass flow	(g/s)	8.33
Specific heat density	$(MW/m^3/bar)$	12.3
Burner load	(kW)	417
Fuel composition	(mole %) CH_4	97.88
	C_2H_6	0.76
	C_3H_8	0.37
	CO_2	0.06
	N_2	0.93

equilibrium chemistry (e.g., presumed pdf using flamelets) completely fail, as they predict combustion in the nozzle [2].

The incompletely mixed fuel/air jet perpendicularly enters the combustion chamber with high, nonswirled velocity, forming a recirculating annulus around the jet due to mass continuity (see Fig. 1). After a certain penetration depth the jet expands, leading to a strong reduction of the velocity magnitude. Prior to the onset of the chemical reactions at appropriate flow velocity values, mixing is further improved on-the-fly by turbulence. Simultaneously, the jet is further preheated by the recirculating hot flue gases. This process is also supported by the hot combustor walls, as is found by *Vaz* [2]. The influence of the entrainment of the flue gases in the jet is not quantified here.

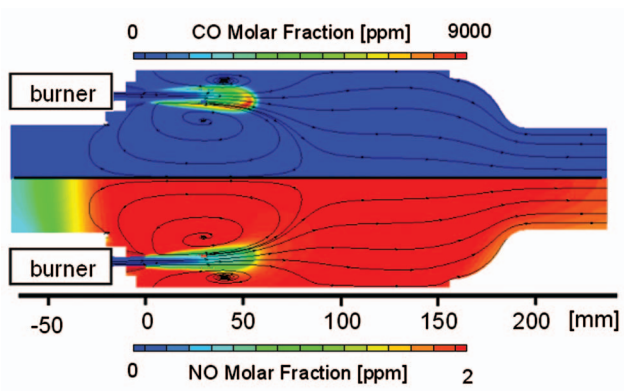

Fig. 1 CO and NO molar fraction distribution and streamline representation of the flow on longitudinal slice through nozzle and combustor axis for λ=2.16. Note: the molar fraction is related to dry gas and 15% O_2 concentration at the combustor exit (see Table 2).

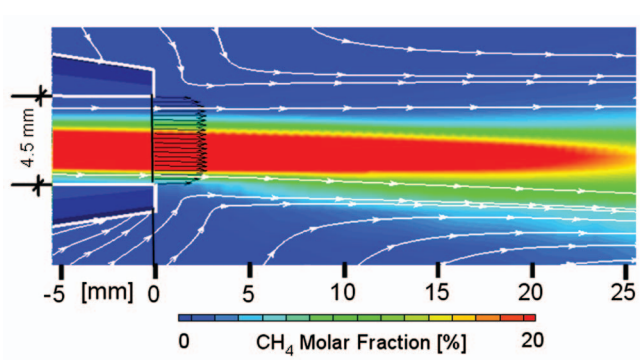

Fig. 2 Inhomogeneous methane distribution at burner nozzle exit, CH_{4min}=0.3%, CH_{4max}=17.7% -turbulent velocity profile at the nozzle exit, $v_{max}\approx$187 m/s, $v_{mean}\approx$173 m/s. Note: Maximum speed of $v\approx$197 m/s is reached about 3.7 mm downstream of the nozzle exit-streamlines indicating recirculation around the jet.

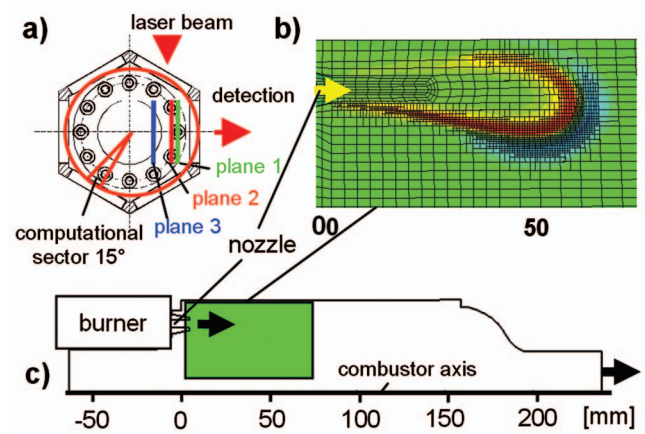

Fig. 3 Schematic of the experimental/computational setup

Due to nonequilibrium chemistry effects the formation of a flame is spatially delayed up to a volumetric heat release at mean temperatures in the range of or even below the adiabatic flame temperature. Accordingly a distinct flame is no more visible. At that temperature the NO-formation is very low (1–10 ppm); far below the equilibrium values which indicates that this process is very slow. In contrast, the CO formation and consumption is very fast, and we may expect exhaust values in the range of the equilibrium data.

Computational and Experimental Setup

The computational domain is derived from the combustor, which was experimentally investigated at the DLR test facility. Combustor length and diameter are 200 and 100 mm, respectively. From Fig. 3(*a*) we see that on the combustor head twelve nozzles (inner Ø 4.5 mm) are equidistantly arranged along a circle (Ø 80 mm) centered about the combustor axis. For a better optical access, the cross section of the combustion chamber on the test rig is of hexagonal shape. However, in the simulations a cylindrical geometry with a combustor diameter of 106 mm is assumed in order to reduce the computational cost. Hence, the computational domain may be limited to a 15 deg sector as indicated in Fig. 3(*a*). The axial extent of the domain is displayed in Fig. 3(*c*). Scalability issues of the investigated FLOX® combustor had not been addressed in the present paper as it was our primary goal to develop and validate our combustion code. However, this topic will be investigated in future work.

For the flow simulation we use the commercial code Fluent 6.1. applying the realizable k-ε model with standard wall functions and the segregated solver. Chemistry and turbulence/chemistry interaction are computed using the DLR combustion code *IGNIS*, which supplies the Fluent code with the source terms for the composition and internal energy transport equations. These transport equations are then solved in Fluent. Based on our experience, the best results concerning performance, stability and accuracy are achieved by performing unsteady computations. Coupling Fluent and *IGNIS* in a steady mode does not converge or is highly inefficient due to the need for very small relaxation factors.

In order to lower computational cost, we use the Fluent code capability to dynamically adapt the computational mesh to the physical solution. Precisely, we start the computation with a very coarse mesh of about 8000 cells. A steady state solution is finally obtained on an adapted mesh of about 80,000 cells. As the adap-

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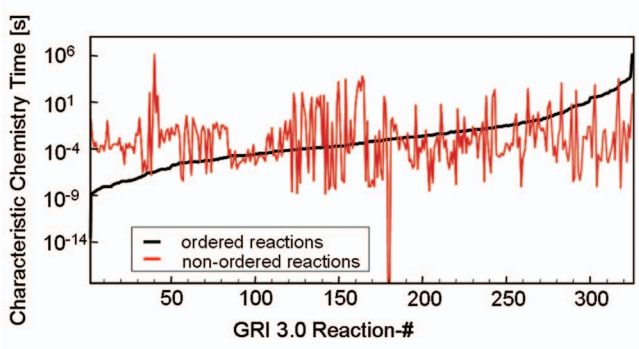

Fig. 4 Characteristic chemistry time over the elementary reactions of the GRI3.0 reaction scheme. Red colored line: nonordered; black colored line: ordered chemical reactions.

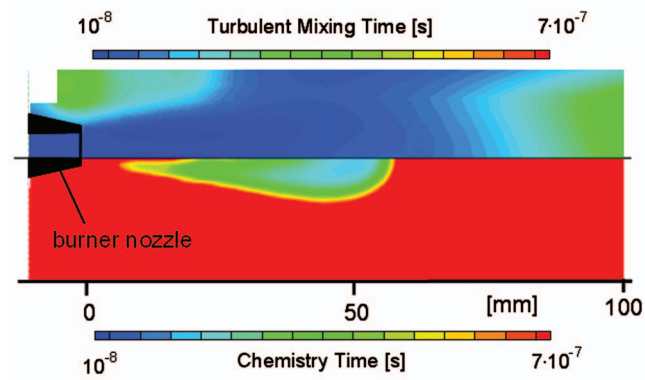

Fig. 5 Characteristic chemistry time (lower half) and turbulent mixing time (upper half) in the combustion zone on a longitudinal slice through the burner nozzle and combustor axis

tation criteria we use the global reaction rate and the production rates for CO and OH. For example a mesh cutout after a mesh-refinement procedure based on the CO production rate is shown in Fig. 3(*b*). Computations are performed for the operational conditions summarized in Table 1.

The Combustion Model

The IGNIS code is a powerful chemistry simulation package incorporating different turbulent combustion models. IGNIS may be coupled with any arbitrary CFD-code via user-defined functions supplying the internal energy and species densities transport equations with the appropriate source terms. The code kernel is a highly optimized routine solving complex chemical systems of elementary reactions applying the "Reference Species Technique" [3] (see Appendix A). This technique is a semi-implicit solution procedure where the individual reaction equations are sequentially solved by linearization in the respective reference species. In this regard we found it crucial for the quality of the solution to rearrange and solve the elementary reactions in the order of increasing characteristic chemistry times. Within the global time step τ the characteristic chemistry time τ_{chem} is then used as the time step in a subcycled explicit solution process of each reaction equation. Thus, the solution of a fast reaction is introduced into the solution process of the next slower reaction, and so on. As an example, the reordering of the reactions in the computational cell of greatest heat release is shown in Fig. 4. While the original order reveals a strongly fluctuating behavior the rearranged order is represented by a smooth curve revealing that the characteristic chemistry times of the fastest and the slowest reaction ranging from 10^{-9} to 10^{6} s, respectively. Unfortunately this order is different for each cell and each time level and has to be computed separately.

With our focus on the pollutant emission we decided to use the GRI3.0 [4] reaction mechanism comprised of 325 elementary reactions and 53 species to investigate the relative importance of the different NO-formation mechanisms. In practice, solution of a very complex chemistry system is prohibitive for the concurrent use of an elaborate turbulence/chemistry interaction model (for example, the presumed pdf approach) by reason of the computational cost. Therefore the simple "Partially Stirred Reactor Model" [5,6] (see Appendix B) is applied in the present computations. This model suffers from the fact that all the turbulent structures in a computational cell have the same mean composition and that there is only one integral turbulent mixing time involved. On this account the model has to be calibrated by adjusting the model constant C_{mix} in Eq. (B5) of Appendix B by comparison with experimental data. For the FLOX®-combustion we found an appropriate value of $C_{\text{mix}}=0.002$. The justification for this will be given later.

In order to evaluate the combustion model it is useful to compute the local turbulent Damköhler number which may be interpreted as a measure of the relative importance of turbulence and chemistry in the combustion zone. From the results of the simulation used below for the code and model validation we first compute the characteristic chemistry time as defined in Eq. (A9) of Appendix A. The distribution of the chemistry time and the characteristic turbulent mixing time is plotted on the lower and upper half of Fig. 5, respectively. Figure 5, as well as subsequent figures, represents a cutout of the reaction zone behind the burner nozzle on a longitudinal slice through burner nozzle and combustor axis.

At first glance we see that the chemistry time is almost one order of magnitude higher than the turbulence mixing time indicating that the FLOX® combustion is notably dominated by finite rate chemistry effects. Both characteristic times are combined to form the Damköhler number defined through Eq. (B10). For comparison the distribution of the heat release rate and the Damköhler number is plotted on the upper and lower half of Fig. 6, respectively.

We observe that heat of combustion is released on an envelope around the impinging fuel/air jet with a maximum of 10^{10} W/m^3 located about 55 mm downstream behind the nozzle exit. We further observe that the Damköhler number strongly correlates with the heat release rate, however, the maximum value of $\text{Da}_T \cong 0.4$ is located slightly upstream of the point of maximum heat release. This fact is also reflected in Fig. 5 by an absolute minimum in the chemistry time.

If the computed Damköhler number is located in the Borghi-

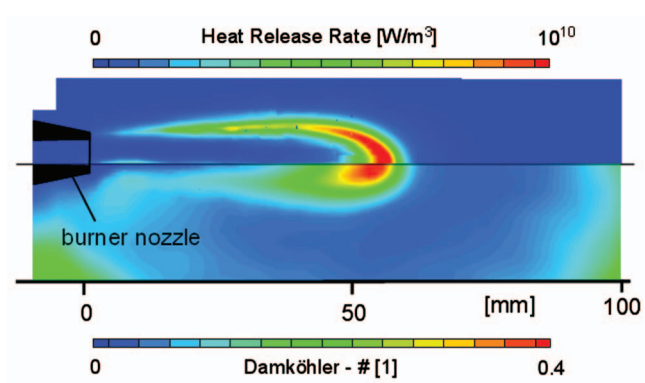

Fig. 6 Damköhler-# (lower half) and heat release rate (upper half) in the combustion zone on a longitudinal slice through the burner nozzle and combustor axis

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Table 2 Comparison of computational and experimental results

Air to fuel ratio λ	2.16				
Max. nozzle velocity (m/s)	195				
Adiabatic temperature (K)	1749				
Equil. CO molar fraction (ppm)	3.8/2.8[a]				
Equil. NO molar fraction (ppm)	2587				
Combustion model	Simulation GRI3.0 quasilaminar		Simulation GRI3.0 PaSR		Experiment
Max. mean temperature (K)	1755		1750		1719[c]
Exhaust temperature (K)	1743		1742		1267[b,d]
Max. CO molar fraction (ppm)	6668		9084		···
Exhaust CO molar fraction (ppm)	3.6	2.3[a]	3.6	2.3[a]	<1.0[a,b]
Max. NO_x molar fraction (ppm)	3.2		3.0		···
Exhaust NO_x molar fraction (ppm)	2.5	1.8[a]	2.3	1.6[a]	6.2[a,b]

[a]Concentration for dry exhaust gas related to 15% O_2 concentration.
[b]Suction thermocouple probing 65 mm behind the combustor exit.
[c]Measured on an area as indicated-in Fig. 9.
[d]Temperature decreased due to undefined amount of cooling air, uncorrected thermocouple.

diagram [7] we find that the FLOX® combustion is characterized by the conditions prevailing in a more or less well stirred reactor.

From the above considerations we may conclude that our choice to use a complex finite rate chemistry scheme along with the partially stirred reactor model is justified in the FLOX® application.

Results

Code and Model Validation. For the validation of the combustion models and the evaluation of the computational results OH-LIF measurement data are available from DLR-experiments relating to the relative amount of OH-radicals and the temperature [8]. Details of the experimental theory are not presented here but may be found in [9]. The species concentrations of CO and NO were detected by suction probing 65 mm downstream of the combustor exit (Ø 50 mm). The outer diameter of the probe is 5 mm and we may assume that the probe does not interfere with the combustor flow. The exhaust temperature was measured with an uncorrected thermocouple. All these values are resumed in Table 2 and used as validation data for the simulations. The measured pressure drop across the combustor of about 5% is coincident with the computational result.

OH. OH is involved in almost all elementary reactions and consequently has an important influence on the accurate prediction of the pollutant formation. In Figs. 7 and 8 the results of the simulations without (quasilaminar) and with application of the PaSR-model are qualitatively compared with the LIF-measured OH-concentration on slice plane 2, as indicated in Fig. 3(*a*).

In the simulations the onset of the OH-formation is rather abrupt with a steep gradient in the reaction zone. In contrast, there is a smooth transition of OH from the zero value in the center of the jets to the surroundings. This is certainly due to a strong turbulent exchange or entrainment across the shear layer between the jets and the recirculating hot flue gases. This physical process is only deficiently described by the k-ε turbulence model. Obviously, this is not only a problem of the combustion model but rather of the turbulence model in the flow simulation code.

The results of the quasilaminar approach presented in Fig. 7 clarifies the pure chemistry aspect of the combustion assuming that the composition at the beginning of each time step is perfectly mixed on the molecular level. This is the limit of infinitely fast turbulent mixing or $\tau_{mix}=0$. Although the flame front is strongly delayed behind the nozzle exit due to finite rate chemistry effects, comparison with the experimental data shows that the reaction is still to fast. This shows that we have to account for the incomplete mixing in the computational cells by application of the PaSR turbulence/chemistry interaction model. As mentioned above the PaSR-model has to be calibrated by adjusting the model parameter C_{mix} in Eq. (B8). For this purpose we used the OH-LIF measurement data to determine a value of $C_{mix}=0.002$ by tuning the conformity of the computational and experimental OH-distribution pattern. The result is shown in Fig. 8.

With the C_{mix} parameter fixed, we further compare computational and experimental results for the temperature and the global emission values at the combustor exit. These values are summarized in Table 2 with reference to the corresponding equilibrium data.

Temperature. Figure 9 shows the temperature distribution on plane 3 of Fig. 3(*a*) related to a color map ranging from 1200 K up to 2000 K. The result of the simulation is displayed on the upper half of the figure while the data from the LIF experiment is shown on a rectangular cutout near the combustor center on the lower half. Comparing both data on the corresponding rectangular areas as outlined in Fig. 9 we find that the computational distribution is almost homogeneous with the average value of 1750 K

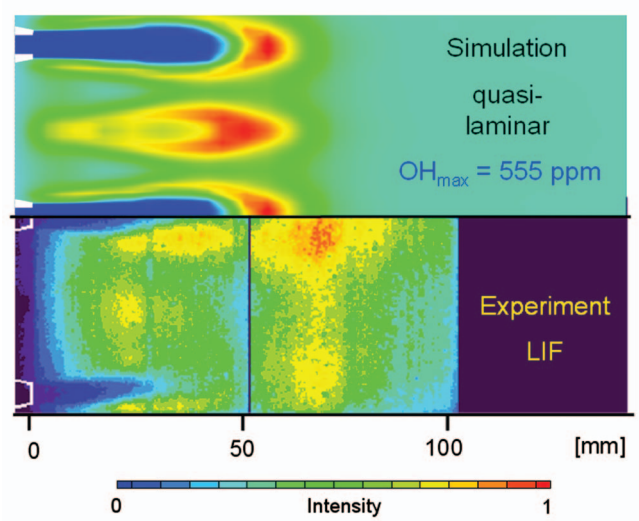

Fig. 7 OH molar fraction distribution on slice plane 2 (see Fig. 3(*a*)) for λ=2.16. Comparison-Simulation: GRI3.0 quasilaminar/ Experiment: LIF.

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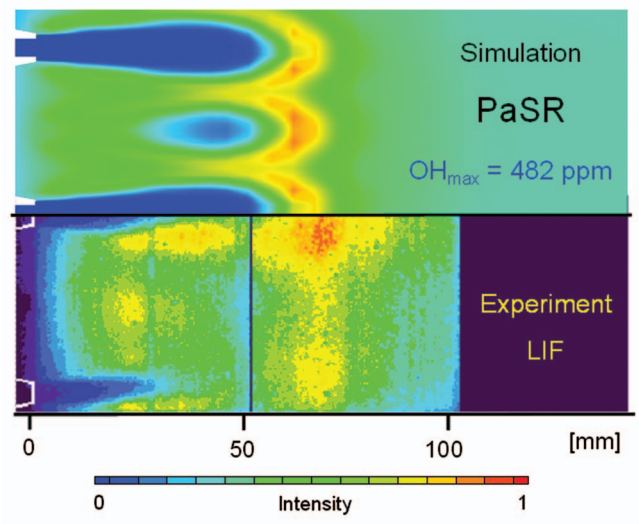

Fig. 8 OH molar fraction distribution on slice plane 2 (see Fig. 3(*a*)) for λ=2.16. Comparison-Simulation: GRI3.0+PaSR-model (C_{mix}=0.002)/Experiment: LIF.

being very close to the adiabatic flame temperature of the global mixture. Against this the experimental data are very inhomogeneous in the range of 1498–1983 K indicating turbulent effects. The averaged value is 1719 K and is about 30 K lower than the computational result. As already discussed above with respect to the OH distribution the computational code is not able to correctly predict this inhomogeneity. The consequence thereof in terms of the NO formation will be discussed below.

Due to some heat loss across the combustor wall the computed exhaust temperature is about 10 K lower than the maximum temperature. This value does not match the experimental value which is almost 475 K lower. However, it has to be noted that in the experiment the exhaust flow is diluted by an unspecified amount of cooling air and that there is no radiative correction of the thermocouple.

CO and NO at the Combustor Exit. The maximum value of the CO molar fraction in the reaction zone is far beyond the equilibrium value of 2.8 ppm. We also find that this value is essentially higher with application of the PaSR model (≈9000 ppm) than with the quasilaminar approach (≈6600 ppm). This tendency is in the opposite direction of the OH formation (see Figs. 7 and 8) which will be investigated in more detail in the next section. However, in both combustion models CO is reduced in the remainder of the combustor and approaches a value of 2.3 ppm related to 15% O_2 at the combustor exit, which is slightly lower than the equilibrium data. The respective experimental value is less than 1.0 ppm.

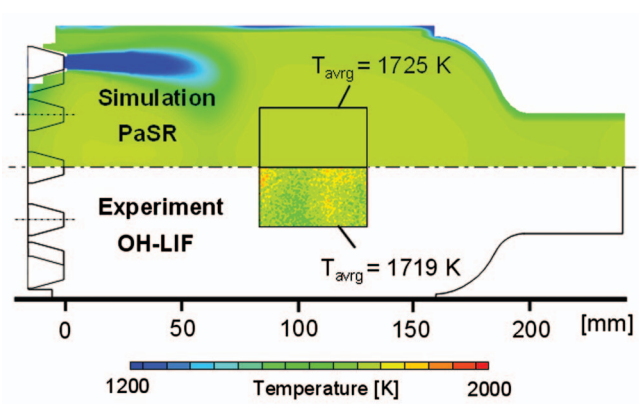

Fig. 9 Temperature distribution on slice plane 3 (see Fig. 3(*a*)) for λ=2.16-Comparison-Simulation: GRI3.0+PaSR/Experiment: OH-LIF

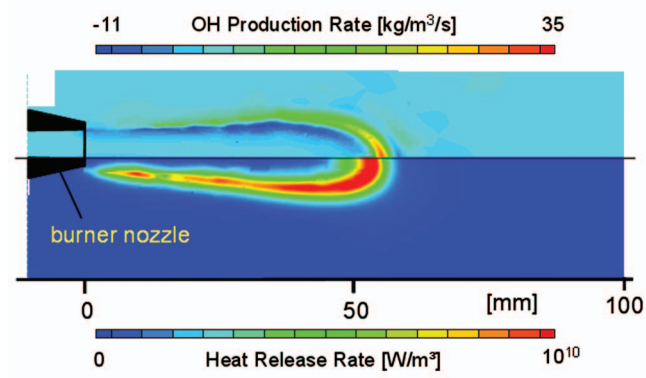

Fig. 10 OH (upper half) production rate and heat release rate (lower half) in the combustion zone on a longitudinal slice through the burner nozzle and combustor axis

Concerning the NO_x formation the computational result at the combustor exit is 1.8 ppm for the quasilaminar model and 1.6 ppm for the PaSR model, respectively. The experimental value is 6.2 ppm. For almost perfectly premixed combustion Leonard and Stegmaier [10] found an NO_x emission value of less than 2 ppm for a combustion temperature of 1750 K. They further found that this is independent of pressure level, preheating temperature, and residence time. Even though mixing on-the-fly prior to combustion—as it is characteristic for the FLOX® combustion—is efficient, we conclude from the work of Leonard and Stegmaier that in the present application the mixing is not yet perfect. One indication of this is given by the measured temperature distribution displayed in Fig. 9 where hot spots of up to 2000 K may be found. The fact is that turbulent structures may contain near-stoichiometric mixture locally resulting in temperature values essentially higher than the cell mean temperature. Accordingly the thermal NO formation in the experiment is higher than the value computed from the mean cell temperature. The reason is, as explained before, that first the k-ε model in the RANS approach has a strong smoothing behavior and second that the PaSR model does not account for different composition in the turbulent substructures of the computational cells. Both deficiencies will be addressed in future works.

In contrast to the decrease of CO, the NO concentration slowly increases towards the combustor exit. Consequently there must be an optimal combustor length with an acceptable level of the CO and NO emission values, as well. The process of CO and NO formation will be considered in more detail in the next section.

Analysis of CO and NO Formation. As mentioned before, OH plays an important role in the formation and consumption of CO as well as NO. Therefore the production rate of OH is incipiently compared with the heat release rate as shown on the upper and lower half of Fig. 10, respectively.

Identifying the layer of maximum heat release with the flame front we observe that the OH production rate increases in the streamwise direction on a cross section through the flame front. Roughly speaking, OH is consumed in front of the flame front and attains its absolute maximum immediately behind the flame front. The formation process of OH is far to complex to be displayed here rather its influence on the pollutant formation is discussed below.

CO Formation. From Fig. 11 we find that the CO formation inversely correlates with the OH formation. For a better under-

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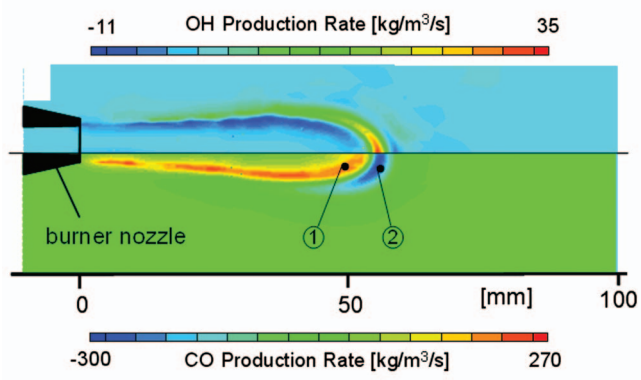

Fig. 11 OH (upper half) and CO (lower half) production rate in the combustion zone on a longitudinal slice through the burner nozzle and combustor axis. 1, 2 analysis points.

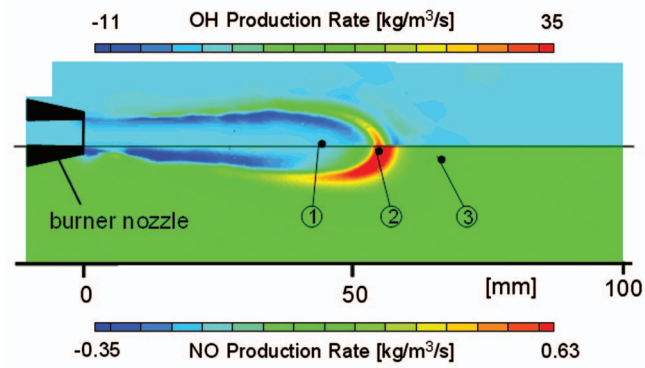

Fig. 13 OH (upper half) and NO (lower half) production rate in the combustion zone on a longitudinal slice through burner nozzle and combustor axis. 1, 2, 3 analysis points.

standing of this fact the GRI3.0 reaction mechanism is analyzed with respect to the CO formation at the two characteristic points of highest production and consumption, as indicated in Fig. 11. The result is depicted in Fig. 12 where the relative amount of the CO production is plotted over the individual equations of the GRI3.0 reaction scheme. The red bars in the charts signify the elementary reactions that produce CO primarily from HCO, viz., $HCO+H_2O \Leftrightarrow CO+H+H_2O$, $HCO+O_2 \Leftrightarrow CO+HO_2$, and $HCO \Leftrightarrow CO+H$. In the blue colored reactions, CO is consumed. We observe that at both analysis points the same elementary reactions are involved. While the CO formation process is comprised of several reactions with several intermediate species involved (see legend in Fig. 11), the CO consumption is primarily described by a single one reaction, viz., $CO+OH \Leftrightarrow CO_2+H$.

At the analysis point 2 where OH is at its maximum this reaction is absolutely predominant over the formation reactions resulting in an almost complete consumption of CO. Against this the CO consumption at the analysis point 1 is essentially diminished for lack of OH radicals. Also the formation reactions are stronger leading to a net production and also to an absolute maximum of CO of about 9000 ppm at this point (see Fig. 1).

NO Formation. In analogy to CO the distribution of the NO formation is compared with the OH formation in Fig. 13. However, in contrast to CO the NO formation directly correlates with the OH formation. Clearly, the production of NO begins with the advent of the OH formation.

Here we analyze the NO formation at the three points indicated in Fig. 13 with the results for the first and second analysis point shown in Fig. 14. The first analysis point is located in front of the reaction zone where we find a weak consumption of NO. The respective analysis of the reaction mechanism shown in Fig. 14 reveals that this primarily results from two competing reactions describing an exchange between NO and NO_2. Of course, this is not really a reduction of the pollutant emission since NO_2 is also a harmful chemical. Additionally there is also a minor amount of NO consumption attributed to Fenimore reactions that are associated with the CH species (see Fig. 14).

The second analysis point is located at the maximum of the NO formation. The respective analysis in Fig. 14 reveals that compared to the first analysis point a lot more reactions are involved. In particular, the number of Fenimore reactions is numerous. These are the blue colored reactions in Fig. 14 and are those reactions that are associated with the CH species (see legend of Fig. 14). In addition we also find a contribution of the three Zeldovich reactions (red) and two reactions describing the NO formation from N_2O (green). The analysis at the third point is similar to that at the second point and is not plotted here.

Recapitulating, the relative amount attributed to the different NO formation mechanisms at the three analysis points and integrated over the whole computational domain is shown in Table 3.

In the reaction zone (point 2) the amount of Fenimore-NO is almost 70% and the amount of NO produced from the Zeldovich, N_2O and NO_2 reactions is approximately in equal measure of

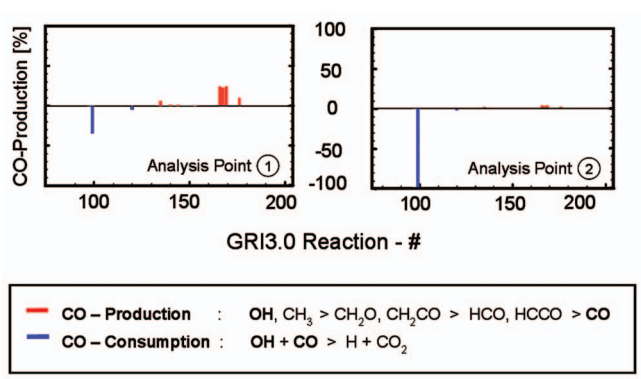

Fig. 12 Relative amount of CO production plotted over the elementary reactions of the GRI3.0 mechanism at analysis point 1 (left) and analysis point 2 (right)

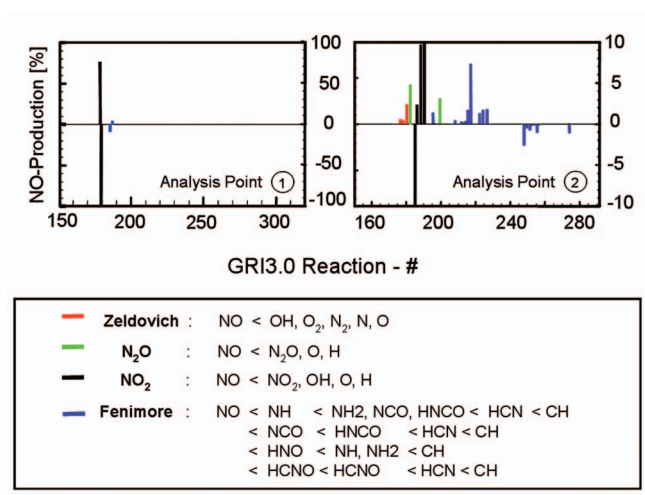

Fig. 14 Relative amount of NO production plotted over the elementary reactions of the GRI3.0 mechanism at analysis point 1 and 2. Red: Zeldovich, Green: N_2O, Black: NO_2, Blue: Fenimore.

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Table 3 Relative amount of the NO formation mechanisms at the three analysis points indicated in Fig. 13 and Integrated over all computational cells (global)

	Point 1	Point 2	Point 3	Global
Zeldovich	0.0%	7.8%	30.5%	40.7%
Fenimore (prompt)	−9.8%	67.3%	33.5%	26.6%
N_2O	0.1%	10.1%	34.2%	31.7%
NO_2	−90.1%	14.8%	−1.8%	−1.0%

about 10%. Beyond the reaction zone at analysis point 3 the Fenimore-NO decreases and simultaneously the relative importance of Zeldovich- and N_2O-NO is increased. NO and NO_2 are almost counterbalanced in the respective reactions. The percentage values at point 3 seem to be representative for the remainder of the combustion chamber. This is evident from the comparison of the global integrated values with the results of point 3. As mentioned above Zeldovich- or thermal NO becomes dominant in the case of high temperatures. Obviously the FLOX®-combustion is characterized by a temperature level where NO produced from the Zeldovich-, Fenimore-, and N_2O-mechanisms is of the same order of magnitude.

Conclusions

In the present paper results of the numerical simulation of the FLOX®-combustion at 20 bar and $\lambda=2.16$ were presented and validated using experimental data from the DLR test facility. For this purpose the highly optimized DLR-combustion code IGNIS was presented and used throughout the computations in a coupled unsteady mode with the commercial code Fluent. The computational results are in good agreement with the experimental data.

In the reaction zone the turbulent Damköhler number is in the range of unity indicating equal importance of the finite rate chemistry and the turbulence/chemistry interaction model. In this regard we successfully applied the GRI3.0 reaction mechanism in combination with the PaSR model. With respect to the NO formation we found that the contribution from the three NO formation mechanisms: Zeldovich, Fenimore, and N_2O is approximately of equal measure. However, in the case of incomplete mixing the PaSR model has to be improved with respect to the inhomogeneous composition in the turbulent substructures within the computational cells. Furthermore it was found, that the turbulent mixing process between the high-velocity fuel/air jet and the recirculating hot flue gases is deficiently described by the k-e turbulence model leaving some potential for further improvement.

Acknowledgment

This work presented was financially supported in part by the European Community within the Project New Combustion Systems for Gas Turbines (NGT) under Contract No. ENK5-CT-2001-00564.

Nomenclature

a_{ir} = stoichiometric coefficients of educt species in reaction r
b_{ir} = stoichiometric coefficients of product species in reaction r
c_i = molar concentration of species i
c_i^r = molar concentration of species i in reaction r
c_i = mean concentration of species i in the cell
C_{mix} = PaSR model constant
Da_T = turbulent Damköhler number
Da_T^r = turbulent Damköhler number defined for reaction r
F_i^r = reaction rate of species i in reaction r
k = turbulent kinetic energy
k_{fr} = Arrhenius coefficients for forward reactions
k_{br} = Arrhenius coefficients for backward reactions
Q = the internal energy
Re_T = turbulent Reynolds number

Greek

ε = dissipation rate of k
Δh_i^0 = heat of formation of species i at absolute zero
κ = reactive volume fraction of the computational cell
ν = kinematic viscosity
τ = infinitesimal time step
τ_{chem}^r = characteristic chemistry time of reaction r
τ_{chem} = integral characteristic chemistry time
τ_{mix} = integral characteristic turbulent mixing time
τ_K = Kolmogorov time scale
τ_T = Taylor time scale

Subscripts

i = species counter
r = elementary reaction counter
K = indicates reference species

Superscripts

r = elementary reaction counter
N = indicates quantity in nonreacting volume fraction of the cell also indicates preceding time level
R = indicates quantity in the reacting volume fraction of the cell

Appendix A: Reference Species Technique

Each combustion process is described by the solution of a system of elementary chemical reactions of the form

$$\frac{dc_i}{dt}=\sum_r \frac{dc_i^r}{dt}=\sum_r F_i^r(c_j) \tag{A1}$$

where c_i is the molar concentration of species i and F_i^r is the rth reaction rate defined as

$$F_i^r(c_j)=(b_{ir}-a_{ir})[\Omega_{\mathrm{fr}}-\Omega_{\mathrm{br}}] \tag{A2}$$

with

$$\Omega_{\mathrm{fr}}:=k_{\mathrm{fr}}\prod_j c_j^{a_{jr}} \quad \text{and } \Omega_{\mathrm{br}}:=k_{\mathrm{br}}\prod_j c_j^{b_{jr}} \tag{A3}$$

.

In Eqs. (A2) and (A3), a_{ir} and b_{ir} are the stoichiometric coefficients and k_{fr} and b_{fr} are the Arrhenius-type reaction coefficients for forward and backward reactions, respectively. Equation (A1) is a strongly nonlinear system of equations whose fully implicit solution, applying the Newton linearization is very tedious. In contrast, although the explicit solution is very simple it is time step dependent and frequently leads to an unphysical composition (negative species densities).

In the IGNIS code, a semi-implicit solution procedure according to the so-called "reference species technique" [3] is applied, which displays a very fast and stable solution behavior. The basic idea is to linearize Eq. (A1) with respect to the species K which for each reaction r is in greatest danger of being driven negative. This species is defined to be the reference species c_K^r following from the condition

$$c_K^r=\left\{c_i^r \,\middle|\, \frac{\frac{dc_i^r}{dt}}{c_i^r}=\text{minimum } \mathbf{i}\right\} \tag{A4}$$

Performing Taylor series expansion of F_i^r around $(F_i^r)^N$ (the upper index N denotes the preceding time level) and neglecting all

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derivatives in the Jacobian matrix except those with respect to the reference species c_K^r leads to the following semi-implicit form of Eq. (A1)

$$\frac{dc_i}{dt} = \sum_r \frac{dc_i^r}{dt} \cong \sum_r (b_{ir} - a_{ir})\dot{\omega}_r^* \tag{A5}$$

with

$$\dot{\omega}_r^* = \frac{\Omega_{\mathrm{fr}}^N - \Omega_{\mathrm{br}}^N}{1 + \dfrac{\tau}{\tau_{\mathrm{chem}}^r}} \tag{A6}$$

where τ_{chem}^r is the characteristic chemistry time for each elementary reaction, viz.,

$$\tau_{\mathrm{chem}}^r := \frac{c_K^{rN}}{(a_{Kr} - b_{Kr})(a_{Kr}\Omega_{\mathrm{fr}}^N - b_{Kr}\Omega_{\mathrm{br}}^N)} \tag{A7}$$

and τ is an infinitesimal time step used to approximate the temporal derivative of the reference species.

In analogy to Eq. (A5) the source term in the internal energy equation reads

$$\frac{dQ}{dt} = -\sum_r \sum_i (b_{ir} - a_{ir})\Delta h_i^0 \dot{\omega}_r^* \tag{A8}$$

where Q is the internal energy and Δh_i^0 is the heat of formation of species i at absolute zero.

In order to specify an expression for a characteristic chemistry time τ_{chem} representative for all elementary reactions in a computational cell we found it reasonable to derive this from Eq. (A8) as a heat release weighted average of the form

$$\tau_{\mathrm{chem}} = \frac{\displaystyle\sum_r Q_r \frac{\tau_{\mathrm{chem}}^r}{\tau + \tau_{\mathrm{chem}}^r}}{\displaystyle\sum_r Q_r \frac{1}{\tau + \tau_{\mathrm{chem}}^r}} \tag{A9}$$

with

$$Q_r := \sum_i (b_{ir} - a_{ir})\Delta h_i^0 (\Omega_{\mathrm{fr}}^N - \Omega_{\mathrm{br}}^N) \tag{A10}$$

Appendix B: Partially Stirred Reactor Model (PaSR)

This model is based upon the fundamental assumption that within a computational cell chemical reaction may occur only in the volume fraction κ of the cell where reactants are mixed on a molecular level. The mean concentration $\overline{c_i}$ of each involved species in the considered cell is then given by the simple linear relation

$$\overline{c_i} = (1 - \kappa)c_i^N + \kappa c_i^R; \quad \kappa \in [0,1] \tag{B1}$$

Here the upper indices R and N denote the reactive and nonreactive part of the cell, respectively. Equation (B1) may be interpreted as a temporal process where c_i^N is the initial concentration and $\overline{c_i}$ is the mean concentration after the lapse of an infinitesimal time τ.

In a turbulent flow, the reactive composition is generated by turbulent dispersion and molecular diffusion. The turbulent dispersion is a very complex process proceeding simultaneously on all turbulent length scales with an inherent spectrum of characteristic mixing times.

In the present model the temporal decay of turbulent eddies up to the Kolmogorov-scale is described by a single integral mixing time τ_{mix} which will be explained later. It is evident that κ is proportional to τ (molecular diffusion) and also proportional to $1 - \tau_{\mathrm{mix}}$ (turbulent dispersion). Assuming equal proportionality Eq. (B1) may be written in the form

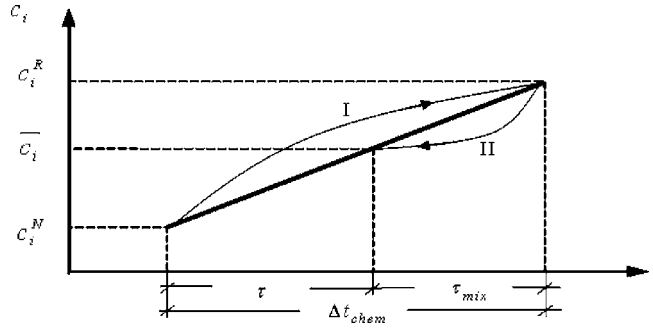

Fig. 15 Graphical representation of Eq. (12)

$$\frac{\overline{c_i} - c_i^N}{\tau} = \frac{c_i^R - \overline{c_i}}{\tau_{\mathrm{mix}}} \equiv F_i^R \tag{B2}$$

Equation (B2) is graphically displayed in Fig. 15 where it is further assumed that the slope of the curve is equal to the reaction rate F_i^R in the reactive part κ of the cell. In Fig. 15 we distinguish two parallel processes I and II:

(I) During reaction time $\Delta t_{\mathrm{chem}} = \tau + \tau_{\mathrm{mix}}$ the initial molar concentration in the reactive fraction κ changes from c_i^N to c_i^R;
(II) The reactive fraction is mixed by turbulence with the nonreactive fraction resulting in the averaged concentration $\overline{c_i}$.

Equation (B2) written for the reference species K of reaction r and Taylor-series expansion of $(F_K^r)^R$ around $\overline{(F_K^r)}$ leads to the implicit form

$$\frac{\overline{(c_K^r)} - (c_K^r)^N}{\tau} = \kappa_r \overline{(F_K^r)} \tag{B3}$$

with

$$\kappa_r := \frac{1}{1 + \mathrm{Da}_T^r} \quad \text{and } \mathrm{Da}_T^r := \frac{\tau_{\mathrm{mix}}}{\tau_{\mathrm{chem}}^r} \tag{B4}$$

In Eq. (B4) a turbulent Damköhler number Da_T^r for reaction r was introduced as the ratio of a characteristic turbulent mixing time related to a characteristic chemistry time.

Semi-implicit formulation of Eq. (B3) using the reference species technique and summing up over all elementary reaction finally yields

$$\frac{d\overline{c_i}}{dt} \cong \frac{\overline{c_i} - c_i^N}{\tau} = \sum_r \kappa_r (b_{ir} - a_{ir})\dot{\omega}_r^* \tag{B5}$$

with

$$\dot{\omega}_r^* = \frac{\Omega_{\mathrm{fr}}^N - \Omega_{\mathrm{br}}^N}{1 + \kappa_r \dfrac{\tau}{\tau_{\mathrm{chem}}^r}} \tag{B6}$$

Comparing Eqs. (B5) and (B6) with Eqs. (A5) and (A6) reveals that the turbulence/chemistry interaction is accounted for through the term κ_r, which depends on the turbulent Damköhler number as specified in Eq. (B4). τ_{chem}^r is given by Eq. (A7) and with definition of the Kolmogorov time scale $\tau_K \sim \sqrt{\nu/\varepsilon}$, the Taylor time scale $\tau_T \sim k/\varepsilon$, and the turbulent Reynolds number $\mathrm{Re}_T := k^2/\upsilon\varepsilon = (\tau_T/\tau_K)^2$ the following expressions for τ_{mix} are found in the literature.

According to the model of Byggstøyl and Magnussen [11]

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$$\tau_{\mathrm{mix}} = \frac{1}{23.6}\left(\frac{k^2}{\upsilon\varepsilon}\right)^{1/4}\frac{k}{\varepsilon} = \frac{1}{23.6}(\mathrm{Re}_T)^{1/4}\tau_T \tag{B7}$$

τ_{mix} increases with increasing turbulent Reynolds number which results in the unphysical fact that the amount of the reactive volume fraction κ decreases.

An alternative approach is given by Karlsson and Chomiak [12] as the geometrical mean of the Kolmogorov and the Taylor time scales, viz.,

$$\tau_{\mathrm{mix}} = C_{\mathrm{mix}}\sqrt{\tau_T\tau_K} = C_{\mathrm{mix}}(\mathrm{Re}_T)^{-(1/4)}\tau_T \tag{B8}$$

Kjäldman [12] reported that best results were obtained by simply using the Taylor time scale

$$\tau_{\mathrm{mix}} = C_{\mathrm{mix}}\tau_T \tag{B9}$$

C_{mix} is a model constant which lies in the range of $C_{\mathrm{mix}}=0.001-0.3$ depending on the flow considered.

For a better interpretation of the computational results it is useful to define global expressions for the fine structure volume fraction κ and the turbulent Damköhler number representative for all reactions in each computational cell. Using Eq. (A9) we may define analogously to Eq. (B4),

$$\mathrm{Da}_T := \frac{\tau_{\mathrm{mix}}}{\tau_{\mathrm{chem}}} \tag{B10}$$

and

$$\kappa := \frac{1}{1+\mathrm{Da}_T} \quad \text{or } \mathrm{Da}_T := \frac{1-\kappa}{\kappa} \tag{B11}$$

Finally, we recognize from Eq. (B11) that within the PaSR-model the Damköhler number may be interpreted as the ratio of the nonreacting, coarse-structured volume fraction related to the reacting, fine-structured volume fraction of each computational cell.

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